

GREIFSWALD, Germany

WMO: 101840

Lat: 54.0967N

Lon: 13.4056E

Elev: 21

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.9	15.2	4.2	6.8	11.0	10.0	9.2	16.2	29.5	40.5	25.3	41.5	3.3	240	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	81.1	66.2	77.2	64.7	73.9	63.1	67.9	77.6	66.0	74.6	64.3	72.0	8.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.5	91.0	71.8	62.6	84.8	69.7	60.8	79.8	68.3	32.2	77.8	30.8	75.1	29.4	72.1	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.2	17.7	8.1	89.5	11.1	4.1	0.2	92.4	-6.3	94.8	-12.5	97.1	-20.5	100.0 (4)		
(5)			7.5	73.2	10.8	2.1	-0.3	74.7	-6.6	75.9	-12.7	77.1	-20.5	78.7 (5)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	33.0	32.4	38.8	45.0	53.4	59.2	63.3	63.0	56.6	49.3	40.2	35.5	(6)
(7)		DBStd	12.78	9.77	8.66	7.22	6.05	5.77	5.21	4.55	4.76	4.47	6.12	5.91	6.50	(7)
(8)		HDD50	2410	528	494	349	175	28	1	0	0	3	86	297	449	(8)
(9)		HDD65	6452	993	912	811	601	363	187	87	98	255	487	745	913	(9)
(10)		CDD50	1524	0	1	2	24	134	278	413	404	202	64	2	0	(10)
(11)		CDD65	93	0	0	0	0	3	14	35	37	4	0	0	0	(11)
(12)		CDH74	735	0	0	0	4	30	122	293	257	28	1	0	0	(12)
(13)		CDH80	159	0	0	0	0	2	24	67	62	4	0	0	0	(13)
(14)	Wind	WSAvg	8.4	10.1	8.8	9.1	8.6	7.6	7.4	7.7	7.1	7.9	8.1	8.9	9.6	(14)
(15)	Precipitation	PrecAvg	23.0	1.7	1.3	1.5	1.4	2.0	2.4	2.6	2.5	2.0	1.9	1.9	1.9	(15)
(16)		PrecMax	33.4	3.3	2.9	3.7	4.1	4.5	5.9	10.8	10.6	5.2	4.3	4.8	4.3	(16)
(17)		PrecMin	14.8	0.0	0.0	0.3	0.1	0.1	0.0	0.0	0.2	0.4	0.4	0.2	0.4	(17)
(18)		PrecStd	4.1	0.8	0.6	0.7	0.8	0.9	1.3	1.7	1.6	1.1	1.0	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	57.2	60.6	70.2	78.0	83.7	85.6	86.2	77.7	69.0	56.0	52.9	(19)
(20)			MCWB	49.2	51.4	51.7	56.3	63.2	66.0	68.1	67.3	65.0	60.8	51.6	49.3	(20)
(21)		2%	DB	48.2	51.0	54.6	63.4	72.8	77.1	81.5	80.4	72.2	64.0	52.5	50.5	(21)
(22)			MCWB	45.4	46.6	48.0	52.4	60.3	64.7	66.8	66.0	62.3	57.7	49.4	47.9	(22)
(23)		5%	DB	46.4	46.1	50.8	58.7	67.9	73.4	76.8	76.1	67.6	60.7	50.5	48.0	(23)
(24)			MCWB	44.1	42.6	45.7	50.0	57.7	62.8	64.9	64.4	59.8	55.2	47.8	45.8	(24)
(25)		10%	DB	44.3	43.1	48.0	54.9	64.1	69.8	73.3	72.3	64.5	57.9	48.2	45.0	(25)
(26)			MCWB	41.7	40.2	44.0	48.3	55.7	60.3	63.2	62.7	57.9	53.4	46.1	43.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	51.4	52.5	57.4	64.2	69.7	70.7	70.3	66.1	61.9	52.5	50.2	(27)
(28)			MCD B	51.3	55.5	60.0	70.1	75.7	77.8	80.8	81.8	76.0	67.1	54.7	52.4	(28)
(29)		2%	WB	46.1	46.5	48.8	53.4	61.4	66.2	68.0	67.4	63.4	58.7	50.0	48.1	(29)
(30)			MCD B	47.5	50.3	53.1	61.6	70.5	74.4	78.3	76.7	70.3	63.2	52.3	50.2	(30)
(31)		5%	WB	44.3	43.0	46.4	51.1	59.0	63.6	66.1	65.6	61.0	56.1	48.2	45.9	(31)
(32)			MCD B	46.3	45.6	50.3	57.6	66.6	71.5	74.1	74.3	66.0	59.8	50.2	47.8	(32)
(33)		10%	WB	41.9	40.3	44.1	48.9	56.5	61.3	64.2	63.7	58.8	54.0	46.3	43.3	(33)
(34)			MCD B	44.1	42.8	47.8	54.3	62.9	68.4	71.5	70.8	63.5	56.9	48.1	45.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.2	10.9	13.9	15.6	15.3	15.9	13.9	12.0	8.3	7.3	(35)	
(36)			MCDBR	8.0	11.3	15.0	20.0	22.7	22.3	23.3	22.4	19.2	15.2	10.1	9.3	(36)
(37)			MCWBR	7.2	8.9	10.3	11.8	12.5	11.9	11.8	11.3	11.2	10.1	8.9	8.5	(37)
(38)		5% WB	MCD BR	8.1	11.5	14.0	18.6	20.4	20.2	20.6	19.9	18.0	13.9	9.7	9.2	(38)
(39)			MCWBR	7.8	9.5	10.3	11.5	12.3	11.6	11.7	10.8	11.4	9.8	8.9	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.341	0.372	0.399	0.396	0.393	0.413	0.411	0.378	0.363	0.340	0.311	(40)	
(41)		taud	2.316	2.312	2.263	2.257	2.310	2.345	2.308	2.324	2.396	2.390	2.359	2.308	(41)	
(42)		Ebn at Noon	200	235	253	262	270	272	263	256	250	225	190	176	(42)	
(43)		Edh at Noon	18	26	33	38	38	37	38	35	29	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	168	368	763	1307	1657	1766	1636	1348	973	508	212	117	(44)	
(45)		RadStd	20	60	75	162	186	132	202	101	92	70	25	11	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (55 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	-281	+103	N/A	(47)

Nomenclature: See separate page